Thyroid Disorders, Osteoporosis and Obesity

All key points for the exam (ATA, AACE, Endocrine Society, BHOF, ACR guidelines) — drug table after each topic

PART A — THYROID DISORDERS

1. Thyroid Physiology and Lab Tests

Hypothalamic–pituitary–thyroid axis

- **TRH** (hypothalamus) → **TSH** (pituitary) → thyroid makes **T4** (~80–90%) and **T3** (~10–20%) → **negative feedback** on TRH and TSH.
- **T4 = prohormone** (long half-life ~**7 days**) • **T3 = active** hormone (half-life ~1 day) • most T3 comes from **peripheral deiodination** of T4 (5'-deiodinase).
- **> 99% protein bound** (mainly **TBG**) • only **free** hormone is active → measure **free T4**.
- Iodine is required: thyroid peroxidase (**TPO**) oxidizes iodide and couples tyrosines → target of **thionamides**.
- ↑ **TBG**: estrogen, pregnancy, tamoxifen, hepatitis, methadone → total T4 ↑ but free T4 normal • ↓ TBG: androgens, glucocorticoids, nephrotic syndrome, cirrhosis.

Interpreting thyroid function tests

Pattern	TSH	Free T4	Meaning
Primary hypothyroidism	↑	↓	Thyroid gland failure (Hashimoto's — most common in iodine-sufficient areas)
Subclinical hypothyroidism	↑	Normal	Early gland failure
Primary hyperthyroidism	↓	↑ (and/or ↑ T3)	Graves', toxic nodule(s), thyroiditis
Subclinical hyperthyroidism	↓	Normal	Mild excess
Central (secondary) hypothyroidism	↓ or inappropriately normal	↓	Pituitary/hypothalamic disease → **monitor with free T4, not TSH**
Nonthyroidal illness (sick euthyroid)	Normal/↓	Normal/↓ • ↓ **T3**	Acute illness — **do not treat**; recheck after recovery

- **TSH is the best single screening test** for primary thyroid disease (log-linear relationship with free T4).
- **Anti-TPO antibodies** → Hashimoto's • **TSI / TRAb** (TSH receptor antibodies) → **Graves'**.
- **Radioactive iodine uptake (RAIU): high** in Graves' and toxic nodules • **low** in thyroiditis, exogenous thyroid hormone, iodine excess.
- **Biotin** (high-dose supplements) interferes with immunoassays → falsely low TSH, falsely high T4 → stop 2 days before testing.
- TSH lags behind free T4 changes by **6–8 weeks** → recheck labs no sooner than **4–8 weeks** after a dose change.

2. Hypothyroidism

Causes

- **Hashimoto's thyroiditis** (chronic autoimmune) — most common • **anti-TPO** positive • women, family history, other autoimmune diseases.
- **Iatrogenic:** thyroidectomy, radioactive iodine, neck radiation.
- **Iodine deficiency** — most common cause worldwide.
- **Drugs: amiodarone**, **lithium**, interferon-alpha, **immune checkpoint inhibitors**, **tyrosine kinase inhibitors** (sunitinib), thionamides (overtreatment), iodine/iodinated contrast.
- Thyroiditis (late hypothyroid phase), central (pituitary/hypothalamic), congenital (newborn screening).

Clinical features

- **Fatigue**, **cold intolerance**, **weight gain**, **constipation**, dry skin, hair loss, brittle nails, depression, impaired memory.
- **Bradycardia**, diastolic hypertension, menorrhagia/infertility, muscle cramps, **delayed relaxation of deep tendon reflexes**, puffy face, hoarseness, goiter.
- **Labs:** ↑ LDL cholesterol, ↑ CK, **hyponatremia**, normocytic/macrocytic anemia.

Treatment — levothyroxine (T4) = drug of choice

- **Full replacement dose: 1.6 mcg/kg/day** (ideal body weight) in young healthy adults.
- **Elderly (> 60–65) or CAD:** start **low** — **12.5–25 mcg/day**, increase by 12.5–25 mcg every 4–6 weeks (avoid angina/arrhythmia).
- **Subclinical / mild:** start 25–75 mcg/day.
- **Goal:** TSH in normal range (≈ 0.5–2.5 young; higher targets 4–6 acceptable in elderly ≥ 70–80).
- **Monitoring:** TSH **every 4–8 weeks** after starting or a dose change → once stable every **6–12 months**. **Central hypothyroidism → monitor free T4**.
- Symptom improvement in **2–3 weeks**; full effect takes weeks to months.
- **Administration:** on an **empty stomach**, **30–60 min before breakfast** (or at bedtime ≥ 3 h after last meal), with water • consistent timing.
- **Narrow therapeutic index** → keep the same manufacturer/product when possible; recheck TSH 6 weeks after switching products.
- **IV dose ≈ 50–75% of oral dose** (e.g., NPO patients; can hold oral for several days because half-life is 7 days).

Separate levothyroxine by ≥ 4 hours from: **calcium**, **iron**, **antacids** (Al/Mg), **sucralfate**, **bile acid sequestrants** (cholestyramine, colesevelam), **phosphate binders** (sevelamer, lanthanum), orlistat, soy, high-fiber food, coffee (30–60 min). **PPIs/H2RAs** ↓ absorption (needs acid). **Enzyme inducers** (rifampin, carbamazepine, phenytoin, phenobarbital) → ↑ dose needed. **Estrogen / pregnancy** → ↑ TBG → ↑ dose needed. **Warfarin:** correcting hypothyroidism ↑ catabolism of clotting factors → ↑ INR → monitor.

Other thyroid preparations

- **Liothyronine (T3, Cytomel)**: short half-life → fluctuating levels, palpitations • not routine; used in myxedema coma (adjunct) or thyroid cancer withdrawal.
- **Desiccated thyroid (Armour)** and **liotrix**: **not recommended** (variable T4:T3 ratio, excess T3).
- **Not** for weight loss or obesity (boxed warning) — large doses may cause serious toxicity.

Subclinical hypothyroidism

- Treat if **TSH > 10 mIU/L**.
- TSH 4.5–10: consider if symptomatic, **positive anti-TPO**, goiter, **pregnancy or planning pregnancy**, infertility, CVD risk, young age.
- Generally **do not treat** mildly elevated TSH in the elderly (TSH rises with age).

Hypothyroidism in pregnancy

- Untreated → miscarriage, preeclampsia, preterm birth, **impaired fetal neurodevelopment** (fetus depends on maternal T4 in 1st trimester).
- **Increase levothyroxine by ~25–30% as soon as pregnancy is confirmed** (e.g., take **2 extra doses per week** = 9 doses/week).
- Goal **TSH < 2.5 mIU/L** (trimester-specific range) • check TSH **every 4 weeks** until mid-pregnancy, then at least once around 30 weeks.
- Return to the pre-pregnancy dose **after delivery**, check TSH ~6 weeks postpartum.

Myxedema coma (medical emergency)

- Severe decompensated hypothyroidism: **altered mental status**, **hypothermia**, bradycardia, hypotension, hypoventilation (↑ CO_2), **hyponatremia**, hypoglycemia.
- **Triggers:** infection, cold exposure, surgery, trauma, **sedatives/opioids**, stopping levothyroxine • elderly women in winter.
- **1. IV levothyroxine:** loading dose **200–400 mcg IV**, then **50–100 mcg IV daily** (or 1.6 mcg/kg × 0.75) until oral possible • lower dose in elderly/cardiac.
- ± **IV liothyronine** 5–20 mcg load then 2.5–10 mcg q8h (adjunct).
- **2. Hydrocortisone 100 mg IV q8h** — give **before/with** thyroid hormone until adrenal insufficiency is excluded (thyroid hormone ↑ cortisol clearance → adrenal crisis).
- **3. Supportive:** mechanical ventilation, passive rewarming (avoid active rewarming → vasodilation/shock), fluids, treat hyponatremia, glucose, treat the infection.

3. Hyperthyroidism and Thyrotoxicosis

Thyrotoxicosis = clinical state of excess thyroid hormone (any cause) • **Hyperthyroidism** = excess production by the thyroid gland.

Causes

Cause	Key features	RAIU	Treatment
Graves' disease (most common)	**TSI/TRAb** stimulate TSH receptor • diffuse goiter, **ophthalmopathy** (exophthalmos), **pretibial myxedema**, young women	**↑ diffuse**	Thionamide, RAI, or surgery + beta-blocker
Toxic multinodular goiter	Elderly, nodular goiter	↑ patchy	**RAI or surgery** (rarely remits)
Toxic adenoma (Plummer's)	Single autonomous nodule	↑ single "hot" nodule	RAI or surgery
Thyroiditis (subacute/painful, painless, **postpartum**)	Release of stored hormone → transient thyrotoxicosis → hypothyroid → recovery • subacute: **painful** neck after viral illness, ↑ ESR	**↓ low**	**Beta-blocker** ± NSAIDs/prednisone (painful) • **no thionamides**
Exogenous (excess levothyroxine)	Low thyroglobulin	↓ low	Reduce dose
Iodine-induced (Jod-Basedow), **amiodarone**	After contrast, amiodarone	↓ low	See amiodarone section

Clinical features

- **Heat intolerance**, sweating, **weight loss** despite ↑ appetite, **palpitations/tachycardia**, **atrial fibrillation** (elderly), tremor, anxiety, insomnia.
- Diarrhea/frequent stools, **oligomenorrhea**, proximal muscle weakness, hyperreflexia, lid lag, warm moist skin, fine hair.
- **Osteoporosis** (↑ bone turnover) and **heart failure** with long-standing disease.
- **Apathetic hyperthyroidism** in elderly: weight loss, AF, depression — few adrenergic signs.

Treatment options for Graves' disease

1. Beta-blockers (symptom control — all patients)

- **Propranolol** 10–40 mg TID–QID (also **blocks peripheral T4 → T3 conversion** at high doses) • alternatives: atenolol, metoprolol, nadolol.
- If beta-blocker contraindicated (asthma): **non-DHP CCB** (diltiazem, verapamil) for rate control.
- Do not treat the underlying disease — taper once euthyroid.

2. Thionamides — methimazole (MMI) and propylthiouracil (PTU)

- **MOA:** inhibit **thyroid peroxidase** → block synthesis of new hormone (do not affect stored hormone → takes **4–8 weeks** to become euthyroid). **PTU also inhibits peripheral T4 → T3 conversion**.
- **Methimazole = preferred** (once daily, less hepatotoxicity): **10–30 mg once daily** (by severity of free T4) → maintenance 5–10 mg/day.
- **PTU only for: 1st trimester of pregnancy**, **thyroid storm**, and patients with minor reactions to MMI who refuse RAI/surgery • dose **50–150 mg TID**.
- **Duration: 12–18 months** then taper/stop if TRAb negative (remission ~30–50%; higher relapse with large goiter, high TRAb, smokers).
- **Monitoring:** free T4 (± T3) every **4–6 weeks** until euthyroid (TSH may stay suppressed for months) • baseline CBC with differential and LFTs.

Thionamide ADRs: agranulocytosis (both; usually first 3 months, dose-related for MMI) → patients must **stop the drug and get a CBC immediately if fever or sore throat**. **Hepatotoxicity: PTU = boxed warning for severe hepatocellular injury/liver failure** • MMI → cholestatic. **Rash/urticaria, arthralgia** (most common minor), ANCA vasculitis (PTU), **MMI teratogenic** in 1st trimester (aplasia cutis, choanal/esophageal atresia). Cross-sensitivity ~50% → do not switch after a major reaction (agranulocytosis, hepatitis).

3. Radioactive iodine (I-131)

- Concentrated in the thyroid → destroys tissue over 6–18 weeks • **most common definitive treatment in the US** for Graves' and toxic nodules.
- Most patients become **hypothyroid** → lifelong levothyroxine (check TFTs 4–8 weeks after).
- **Contraindicated: pregnancy** and **breastfeeding**, active moderate-severe ophthalmopathy, thyroid cancer suspicion • avoid pregnancy for **6 months** after.
- Pre-treat with MMI in elderly/cardiac patients → **stop MMI 3–5 days before RAI**.
- Can **worsen Graves' ophthalmopathy** → give **prednisone** prophylaxis in at-risk patients; smoking cessation.
- Transient radiation thyroiditis → may worsen thyrotoxicosis (beta-blocker).

4. Thyroidectomy

- For large goiters, compressive symptoms, suspected cancer, severe ophthalmopathy, pregnancy (2nd trimester) if drugs fail.
- Pre-op: make euthyroid with MMI + **potassium iodide (SSKI or Lugol's) for 7–10 days before surgery** (Graves') → ↓ vascularity and hormone release.
- Complications: **hypoparathyroidism → hypocalcemia**, recurrent laryngeal nerve injury (hoarseness), hypothyroidism.

Iodides (SSKI, Lugol's solution)

- **Wolff–Chaikoff effect**: high iodine acutely **blocks hormone release and synthesis** → fastest drop in hormone levels • effect escapes after 1–2 weeks.
- Uses: thyroid storm, pre-thyroidectomy, radiation emergency (KI protects thyroid).
- **Give iodine ≥ 1 hour AFTER the thionamide** in storm (otherwise iodine becomes substrate for new hormone).
- ADRs: metallic taste, hypersensitivity, salivary gland swelling, acneiform rash.

Thyroid storm (thyroid crisis) — emergency

- Life-threatening thyrotoxicosis: **high fever**, **tachycardia**/AF, heart failure, agitation/delirium, coma, vomiting/diarrhea, jaundice • Burch–Wartofsky score.

- **Triggers:** infection, surgery, trauma, RAI, iodinated contrast, stopping thionamides, DKA, childbirth.

Step	Drug and dose	Purpose
1. Block synthesis	**PTU 500–1,000 mg load → 250 mg q4h** (preferred: also ↓ T4→T3) **or** MMI 60–80 mg/day divided	Stop new hormone production
2. Block release (≥ 1 h after thionamide)	**SSKI 5 drops q6h** or **Lugol's 4–8 drops q6–8h**	Wolff–Chaikoff effect
3. Block adrenergic effects	**Propranolol 60–80 mg PO q4h** (or IV esmolol)	Rate control, ↓ T4→T3
4. Glucocorticoid	**Hydrocortisone 300 mg IV load → 100 mg q8h** (or dexamethasone)	↓ T4→T3, treats relative adrenal insufficiency
5. Supportive	Cooling, **acetaminophen** for fever (**avoid aspirin/salicylates** → displace T4 from TBG), fluids, treat trigger	
Refractory	Cholestyramine (↓ enterohepatic recycling), plasmapheresis, thyroidectomy	

Hyperthyroidism in pregnancy

- **PTU in the 1st trimester** (MMI teratogenic) → **switch to MMI in the 2nd and 3rd trimesters** (PTU hepatotoxicity).
- Use the **lowest dose** to keep free T4 at the **upper limit of normal** (thionamides cross the placenta → fetal hypothyroidism/goiter).
- **RAI contraindicated** • surgery in the 2nd trimester if needed • beta-blockers short term only (fetal growth restriction).
- Gestational transient thyrotoxicosis (hCG stimulates TSH receptor, hyperemesis) → no thionamide.
- **Breastfeeding:** MMI ≤ 20 mg/day (or PTU ≤ 450 mg/day) is acceptable.

Subclinical hyperthyroidism

- Treat if **TSH < 0.1** and: age ≥ 65, cardiac disease, **osteoporosis**, postmenopausal not on estrogen/bisphosphonates, or symptoms (risk: **AF** and **bone loss**).

4. Drug-Induced Thyroid Disease

Drug	Effect	Notes
Amiodarone (37% iodine by weight)	**Hypothyroidism** (more common; continue amiodarone + levothyroxine) • **thyrotoxicosis**: **type 1** (iodine-induced synthesis → thionamide) or **type 2** (destructive thyroiditis → **prednisone**)	TFTs at **baseline and every 3–6 months** • long half-life → effects persist after stopping
Lithium	**Hypothyroidism**, goiter (↓ hormone release) • rarely hyperthyroidism	TFTs baseline then every 6–12 months
Interferon-alpha, **IL-2**	Thyroiditis, hypo- or hyperthyroidism	Monitor TSH
Immune checkpoint inhibitors (pembrolizumab, nivolumab, ipilimumab)	Thyroiditis → hypothyroidism, hypophysitis (central)	TSH every 4–6 weeks
Tyrosine kinase inhibitors (sunitinib)	Hypothyroidism	Monitor TSH
Iodine / iodinated contrast	Hyper (Jod-Basedow, nodular goiter) or hypo (Wolff–Chaikoff)	
Glucocorticoids, dopamine, octreotide	↓ **TSH** secretion	Central effect
Estrogen, tamoxifen, methadone	↑ **TBG** → ↑ levothyroxine requirement	Free T4 normal
Rifampin, phenytoin, carbamazepine, phenobarbital	↑ T4 metabolism → ↑ dose needed	

Drug	Effect	Notes
Biotin	Assay interference	Stop before testing

5. Thyroid Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Levothyroxine (T4)	1.6 mcg/kg/day • elderly/CAD 12.5–25 mcg • IV 50–75% of PO • myxedema 200–400 mcg IV load	Signs of hyperthyroidism (overdose): palpitations, AF, **bone loss**	Empty stomach • **separate 4 h** from Ca, Fe, PPIs ↓, binders • inducers ↑ dose • **pregnancy +30%** • ↑ warfarin effect • NTI
Liothyronine (T3)	5–25 mcg/day • myxedema 5–20 mcg IV	Palpitations, arrhythmia	Not routine
Desiccated thyroid	60 mg ≈ 100 mcg T4	Excess T3	**Not recommended**
Methimazole	**10–30 mg daily** → 5–10 mg maintenance • storm 60–80 mg/day	**Agranulocytosis**, rash, arthralgia, cholestatic hepatitis, **teratogenic (1st trimester)**	**Preferred thionamide** • stop 3–5 days before RAI • ↓ warfarin effect as euthyroid
Propylthiouracil	**50–150 mg TID** • storm 500–1,000 mg load → 250 mg q4h	**Hepatotoxicity (boxed)**, agranulocytosis, ANCA vasculitis	**1st trimester** and **thyroid storm** • ↓ T4→T3
Propranolol	10–40 mg TID–QID • storm 60–80 mg q4h	Bronchospasm, bradycardia, fatigue, masks hypoglycemia	↓ T4→T3 conversion
Atenolol / metoprolol	25–100 mg daily	Bradycardia	Once daily option
Diltiazem / verapamil	Per rate control	Bradycardia, constipation	If beta-blocker contraindicated
Potassium iodide (SSKI / Lugol's)	SSKI 5 drops q6h • Lugol's 4–8 drops q6–8h • pre-op 7–10 days	Metallic taste, rash, sialadenitis	**Give ≥ 1 h after thionamide** • Wolff–Chaikoff
Radioactive iodine (I-131)	Single oral dose	**Hypothyroidism**, worsens ophthalmopathy, radiation thyroiditis	**CI pregnancy/breastfeeding**
Hydrocortisone	Storm: 300 mg → 100 mg q8h • myxedema: 100 mg q8h	Hyperglycemia	↓ T4→T3
Prednisone	Type 2 AIT / painful thyroiditis 40 mg/day taper • RAI eye prophylaxis	Hyperglycemia	
Cholestyramine	4 g QID (refractory storm)	Constipation	Binds hormone in gut

PART B — OSTEOPOROSIS

6. Definition, Risk Factors and Diagnosis

- **Low bone mass + microarchitectural deterioration** → ↑ fracture risk • most fractures: **vertebral** (often silent), **hip** (highest mortality), **wrist**.
- Bone remodeling: **osteoclasts resorb**, **osteoblasts form** • after menopause **estrogen loss** → ↑ **resorption** (rapid bone loss first 5–10 years).

Diagnosis — DXA (central: lumbar spine, femoral neck, total hip)

Category	T-score	Notes
Normal	≥ −1.0	
Low bone mass (osteopenia)	−1.0 to −2.5	Use **FRAX** to decide on treatment
Osteoporosis	**≤ −2.5**	Or **hip/vertebral fragility fracture** regardless of BMD • or osteopenia + fragility fracture of humerus, pelvis, forearm
Severe osteoporosis	≤ −2.5 + fragility fracture	

- **T-score** = compared with young healthy adults → postmenopausal women and men ≥ 50.
- **Z-score** = compared with same age/sex → **premenopausal women, men < 50, children** • Z ≤ −2.0 = "below expected range for age" → look for secondary causes.
- **Fragility fracture** = fracture from a fall from standing height or less.
- **Screening:** all **women ≥ 65** • postmenopausal women < 65 with risk factors (FRAX) • **men ≥ 70** (BHOF) or 50–69 with risk factors • adults with fragility fracture or on glucocorticoids.
- **FRAX** = 10-year probability of hip and major osteoporotic fracture (uses femoral neck BMD ± clinical risk factors).
- Height loss ≥ 1.5 in (4 cm) or kyphosis → vertebral imaging.
- Repeat DXA every **1–2 years** after starting therapy, then less often when stable.

Risk factors

- **Non-modifiable:** age, **female sex**, white/Asian race, **family history** (parental hip fracture), **previous fragility fracture** (strongest), early menopause.
- **Modifiable: low body weight** (< 58 kg / BMI < 20), **smoking**, **alcohol ≥ 3 drinks/day**, low calcium/vitamin D, inactivity, falls.
- **Diseases:** RA, hyperthyroidism, **hyperparathyroidism**, hypogonadism, celiac/IBD/malabsorption, gastric bypass, CKD, diabetes, COPD, anorexia, multiple myeloma.

Drugs that cause bone loss / fractures: glucocorticoids (≥ 2.5 mg prednisone ≥ 3 months — most common secondary cause), **excess levothyroxine**, **anticonvulsants** (phenytoin, phenobarbital, carbamazepine — vitamin D metabolism), **heparin** (long term), **medroxyprogesterone depot** (boxed: limit to 2 years), **aromatase inhibitors** (anastrozole, letrozole), **GnRH agonists**/androgen deprivation, **PPIs** (long term), **SSRIs**, **thiazolidinediones**, canagliflozin, loop diuretics, calcineurin inhibitors, methotrexate (high dose). Sedatives/antihypertensives (orthostasis) ↑ **falls**.

Labs to exclude secondary causes

- Calcium, phosphorus, **25-OH vitamin D**, creatinine, alkaline phosphatase, **TSH**, CBC, ± PTH, testosterone (men), celiac antibodies, SPEP, 24-h urine calcium.

7. Prevention and Nonpharmacologic Therapy

- **Calcium (total from diet + supplements): 1,000 mg/day** (women ≤ 50, men ≤ 70) • **1,200 mg/day** (**women > 50, men > 70**) • prefer dietary sources • excess supplements → kidney stones, possible CV risk.
- **Vitamin D: 800–1,000 IU/day** for adults ≥ 50 (RDA 600 IU < 70, 800 IU ≥ 70) • goal **25-OH vitamin D ≥ 30 ng/mL** • deficiency: 50,000 IU weekly × 8 weeks.
- **Weight-bearing** and muscle-strengthening exercise, balance training (tai chi).
- **Fall prevention:** home safety, vision, review sedating drugs, hip protectors.
- **Stop smoking**, limit alcohol (≤ 2 drinks/day), adequate protein.

Calcium salt	% elemental Ca	Administration	Notes
Calcium carbonate	**40%**	**With meals** (needs acid)	Cheapest • constipation, bloating • ↓ absorption with PPIs/achlorhydria

Calcium salt	% elemental Ca	Administration	Notes
Calcium citrate	**21%**	**Any time** (with or without food)	Preferred with **PPIs/H2RAs**, achlorhydria, **gastric bypass** • less constipation
Calcium gluconate / lactate	9% / 13%		Low elemental content

- Max **500–600 mg elemental Ca per dose** (absorption saturates) → divide doses.
- **Calcium chelates: levothyroxine** (4 h), **quinolones**, **tetracyclines**, **iron**, **bisphosphonates** (take bisphosphonate first), dolutegravir/integrase inhibitors.
- Thiazides ↓ urinary calcium (→ hypercalcemia with large supplement doses).

8. Pharmacologic Treatment

Who to treat

- **T-score ≤ −2.5** at spine, femoral neck, or total hip.
- **Hip or vertebral fracture** (clinical or morphometric) regardless of BMD.
- **Osteopenia + FRAX 10-year probability ≥ 20% major osteoporotic fracture or ≥ 3% hip fracture**.

Choosing therapy by risk (AACE 2020 / Endocrine Society)

Risk category	Definition	Initial therapy
High risk	Osteoporosis without very-high-risk features	**Alendronate, risedronate, zoledronic acid, denosumab** • ibandronate/raloxifene if **spine-only** risk
Very high risk	**Fracture within past 12 months**, fracture while on therapy, multiple fractures, fracture on bone-harming drugs (steroids), **T-score < −3.0**, high fall risk, FRAX MOF > 30% / hip > 4.5%	**Anabolic first**: **abaloparatide, teriparatide, romosozumab** (or zoledronic acid/denosumab) → **then follow with an antiresorptive**

- **ACP 2023: bisphosphonates = first-line** for most postmenopausal women and men • denosumab = second-line (if bisphosphonate contraindicated/not tolerated).
- **Bisphosphonates** ↓ vertebral, hip, and nonvertebral fractures (alendronate, risedronate, zoledronic acid) • **ibandronate** → **vertebral only**.
- **Anabolic agents** must be followed by an antiresorptive to keep the gained BMD.

Bisphosphonates (antiresorptive — inhibit osteoclasts)

Drug	Dose	Notes
Alendronate (Fosamax)	Treatment **70 mg weekly** or 10 mg daily • prevention 35 mg weekly / 5 mg daily	Avoid CrCl < 35 • oral solution and effervescent tablet available
Risedronate (Actonel)	**35 mg weekly**, **150 mg monthly**, 5 mg daily	**Atelvia** (DR 35 mg weekly) — take **right after breakfast** • avoid CrCl < 30
Ibandronate (Boniva)	**150 mg PO monthly** or **3 mg IV every 3 months**	**Vertebral fractures only** • stay upright and fast **60 min** • avoid CrCl < 30
Zoledronic acid (Reclast)	**5 mg IV once yearly** (prevention every 2 years) over ≥ 15 min	**CI CrCl < 35** • acute-phase reaction (give acetaminophen) • hydrate • good for adherence/GI intolerance

Oral bisphosphonate administration (very commonly tested): take **first thing in the morning** on an **empty stomach** with a **full glass (6–8 oz) of plain water** (no coffee, juice, mineral water) • stay **upright (sitting/standing) for 30 minutes** (60 min ibandronate) • wait **30 minutes** (60 min ibandronate) before food, drinks, or other meds (especially **calcium, iron, antacids, Mg**) • missed weekly dose → take next morning (if not same day as next dose). **Correct hypocalcemia and vitamin D deficiency before starting**.

- **ADRs: esophagitis/esophageal ulcer**, dyspepsia, **musculoskeletal pain**, **hypocalcemia**, **osteonecrosis of the jaw (ONJ)** (rare; higher with IV oncology doses, dental surgery → dental exam before starting), **atypical femur fractures** (thigh/groin pain → X-ray; long-term use), acute-phase reaction (IV: fever, myalgia), ocular inflammation (uveitis), nephrotoxicity (IV).
- **Contraindications (oral): esophageal abnormalities** (stricture, achalasia), **inability to stand/sit upright 30 min**, hypocalcemia, CrCl < 30–35, aspiration risk.
- **Drug holiday:** consider after **5 years oral** or **3 years IV** if low–moderate risk • continue up to **10 years oral / 6 years IV** if high risk • reassess every 2–3 years.

Denosumab (Prolia) — RANKL monoclonal antibody

- **60 mg SC every 6 months** • ↓ vertebral, hip, and nonvertebral fractures • **no renal dose adjustment** (but see hypocalcemia).
- **ADRs: hypocalcemia** (**boxed warning 2024: severe hypocalcemia in advanced CKD, eGFR < 30 or dialysis** → check Ca/CKD-MBD before), **serious infections** (cellulitis, UTI), dermatitis/eczema, ONJ, atypical femur fractures, hypercholesterolemia.
- **Do not stop or delay doses:** stopping → **rapid bone loss and multiple vertebral fractures** (rebound) → **no drug holiday**; when stopping, **switch to a bisphosphonate** (e.g., zoledronic acid).
- Given by a healthcare professional • also used in men, glucocorticoid-induced osteoporosis, and androgen-deprivation / aromatase inhibitor bone loss.
- **Xgeva** (120 mg monthly) = oncology (bone metastases, hypercalcemia of malignancy) — different product.

Anabolic agents

Drug	Dose	Key points
Teriparatide (Forteo — PTH 1-34)	**20 mcg SC daily** (thigh or abdomen)	**Orthostatic hypotension** (first doses — sit/lie down), **hypercalcemia**, leg cramps, nausea, ↑ uric acid • osteosarcoma boxed warning **removed 2020** (rats) • use > 2 years in lifetime only if still high risk • avoid in **Paget's**, open epiphyses, prior skeletal radiation, bone metastases, unexplained ↑ alk phos • refrigerate pen
Abaloparatide (Tymlos — PTHrP analog)	**80 mcg SC daily**	Similar to teriparatide (orthostasis, hypercalcemia, **hypercalciuria**, palpitations) • max 2 years • no refrigeration after first use (30 days) • approved for men too
Romosozumab (Evenity — sclerostin inhibitor)	**210 mg SC monthly × 12 months** (two 105 mg injections)	**Dual action**: ↑ formation + ↓ resorption • **Boxed: MI, stroke, CV death** → **do not use if MI or stroke in past year** • hypocalcemia, ONJ, AFF • **limit to 12 doses** then antiresorptive

- Sequence: **anabolic → antiresorptive** gives the best BMD gains • switching antiresorptive → teriparatide gives a smaller gain (especially from denosumab: transient hip loss).

Other agents

- **Raloxifene** (Evista, SERM) **60 mg daily**: estrogen **agonist on bone**, **antagonist on breast and uterus** • ↓ **vertebral fractures only** • ↓ **invasive breast cancer risk** • **boxed: VTE** (DVT/PE) and **fatal stroke** (women with CHD) • ADRs: **hot flashes**, leg cramps, peripheral edema • **stop 72 h before prolonged immobilization** • CI: history of VTE, pregnancy • cholestyramine ↓ absorption, ↓ warfarin effect.
- **Bazedoxifene + conjugated estrogens** (Duavee): prevention in women with a uterus + vasomotor symptoms.
- **Estrogen/hormone therapy**: prevention only in women with menopausal symptoms (< 60 years or < 10 years from menopause) — **not first-line for osteoporosis** (WHI: breast cancer, stroke, VTE, CHD).
- **Calcitonin** (nasal 200 units daily, alternate nostrils; SC/IM 100 units): **least effective** (vertebral only) • **analgesic effect for acute painful vertebral fracture** • rhinitis, epistaxis • **malignancy risk** with long-term use → last resort.
- **Testosterone**: men with hypogonadism (adjunct, not alone if high fracture risk).

Special populations

- **Glucocorticoid-induced osteoporosis (ACR 2022):** all patients on ≥ 2.5 mg prednisone ≥ 3 months → calcium + vitamin D + lifestyle • moderate/high risk → **oral bisphosphonate first-line** • alternatives: IV bisphosphonate, denosumab, teriparatide/abaloparatide (very high risk) • **use the lowest steroid dose**.
- **Men:** alendronate, risedronate, zoledronic acid, denosumab, teriparatide, abaloparatide.
- **Premenopausal / pregnancy:** avoid bisphosphonates (long skeletal half-life, teratogenic potential) — treat cause.
- **CKD stage 4–5:** rule out CKD-MBD first; denosumab with close calcium monitoring (boxed) • avoid bisphosphonates if CrCl < 30–35.

9. Osteoporosis Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Calcium carbonate	1,000–1,200 mg/day total (≤ 500–600 mg/dose), **with meals**	Constipation, bloating, kidney stones	PPIs ↓ absorption • chelates levothyroxine, quinolones, tetracyclines, iron, bisphosphonates
Calcium citrate	Same total • any time	Less GI upset	**With PPIs**, gastric bypass
Vitamin D3 (cholecalciferol)	800–1,000 IU/day • deficiency 50,000 IU weekly × 8 weeks	Hypercalcemia (toxicity)	Goal 25-OH D ≥ 30 ng/mL
Alendronate	**70 mg weekly** / 10 mg daily	**Esophagitis**, MSK pain, ONJ, AFF, hypocalcemia	**Upright 30 min, plain water, empty stomach** • avoid CrCl < 35 • holiday after 5 y
Risedronate	35 mg weekly / 150 mg monthly / 5 mg daily	Same	Atelvia after breakfast • avoid CrCl < 30
Ibandronate	150 mg PO monthly • 3 mg IV q3 months	Same	**60 min** rule • **vertebral only**
Zoledronic acid	**5 mg IV yearly**	**Acute-phase reaction**, hypocalcemia, nephrotoxicity, AF, ONJ	**CI CrCl < 35** • hydrate • holiday after 3 y
Denosumab	**60 mg SC every 6 months**	**Hypocalcemia (boxed in advanced CKD)**, infections, dermatitis, ONJ	**No holiday — rebound fractures** → follow with bisphosphonate
Teriparatide	**20 mcg SC daily**	**Orthostatic hypotension**, hypercalcemia, leg cramps	Anabolic • avoid Paget's/radiation • ↑ digoxin toxicity with hypercalcemia
Abaloparatide	**80 mcg SC daily**	Orthostasis, hypercalcemia, hypercalciuria	Max 2 years
Romosozumab	**210 mg SC monthly × 12**	**MI, stroke, CV death (boxed)**, hypocalcemia, ONJ	No MI/stroke in past year
Raloxifene	**60 mg daily**	**VTE, fatal stroke (boxed)**, hot flashes, leg cramps	↓ breast cancer • vertebral only • stop 72 h before immobilization • cholestyramine ↓
Calcitonin (nasal)	200 units daily (alternate nostrils)	Rhinitis, epistaxis, **malignancy**	Least effective • acute vertebral fracture pain
Conjugated estrogens/bazedoxifene	0.45/20 mg daily	VTE, stroke (boxed for estrogens)	Prevention + vasomotor symptoms

PART C — OBESITY

10. Definition, Assessment and Complications

Category	BMI (kg/m²)	Asian populations
Underweight	< 18.5	< 18.5
Normal	18.5–24.9	18.5–22.9
Overweight	**25–29.9**	23–24.9 (or 23–27.4)
Obesity class I	**30–34.9**	**≥ 25** (or ≥ 27.5)
Obesity class II	**35–39.9**	
Obesity class III (severe)	**≥ 40**	

- **BMI = weight (kg) / height (m)²** • does not distinguish fat from muscle → also measure **waist circumference**.
- **Waist circumference (central/visceral obesity)**: **men > 40 in (102 cm)**, **women > 35 in (88 cm)** (Asian: men ≥ 90 cm, women ≥ 80 cm) → ↑ cardiometabolic risk.
- **Children:** overweight **BMI ≥ 85th** percentile • obesity **≥ 95th** percentile • severe obesity ≥ 120% of the 95th percentile.
- Obesity is a **chronic disease** • goal: **5–10% weight loss** → clinically meaningful improvement in BP, glucose, lipids, OSA, knee OA, MASLD.

Obesity-related complications

- **Type 2 diabetes**, prediabetes, **hypertension**, dyslipidemia (↑ TG, ↓ HDL), **ASCVD**, heart failure (HFpEF), AF, **obstructive sleep apnea**, **MASLD/MASH** (fatty liver), **osteoarthritis**, GERD, gallstones, PCOS/infertility, cancers (endometrial, breast, colon), depression, VTE.

Secondary causes / weight-gaining drugs (review the medication list first)

- **Diseases:** hypothyroidism, Cushing's syndrome, PCOS, hypothalamic injury, genetic (POMC, leptin receptor, Prader-Willi, Bardet-Biedl).

Drug class	Weight gain	Weight-neutral / loss alternative
Antipsychotics	**Olanzapine, clozapine** > quetiapine, risperidone	Aripiprazole, ziprasidone, lurasidone
Antidepressants	**Mirtazapine**, TCAs, **paroxetine**, MAOIs	**Bupropion**, fluoxetine, sertraline
Anticonvulsants / mood stabilizers	**Valproate**, carbamazepine, **gabapentin, pregabalin**, **lithium**	**Topiramate**, zonisamide, lamotrigine
Diabetes drugs	**Insulin**, **sulfonylureas**, **TZDs**, meglitinides	**Metformin**, **GLP-1 RAs**, **SGLT2i**, DPP-4i (neutral)
Others	**Glucocorticoids**, older beta-blockers (propranolol, metoprolol), **DMPA**, antihistamines, megestrol	ACEi/ARB, carvedilol, nonhormonal or IUD contraception

11. Treatment of Obesity

Lifestyle (foundation for everyone)

- **Reduced-calorie diet:** deficit of **500–750 kcal/day** (≈ 1,200–1,500 kcal/day women, 1,500–1,800 men) → ~0.5–1 kg/week loss.
- **Physical activity: ≥ 150 min/week** moderate aerobic activity (200–300 min/week for weight maintenance) + resistance training.
- **Behavioral therapy:** self-monitoring, goal setting, ≥ 14 sessions in 6 months.

Indications for pharmacotherapy (adjunct to lifestyle)

BMI ≥ 30 or BMI ≥ 27 with ≥ 1 weight-related comorbidity (HTN, T2DM, dyslipidemia, OSA, CVD). **Assess response at 12–16 weeks on the maintenance dose:** if **< 5% weight loss → stop** and try another option. **All weight-loss drugs are contraindicated in pregnancy** (weight loss offers no benefit). Weight regain is expected after stopping → long-term therapy.

Anti-obesity medications

Drug	Mechanism	Typical weight loss
Tirzepatide (Zepbound)	**GIP + GLP-1** receptor agonist	**~20%** (most effective)
Semaglutide (Wegovy)	GLP-1 RA	**~15%**
Phentermine/topiramate ER (Qsymia)	Sympathomimetic (↓ appetite) + GABA/glutamate (satiety)	~9–10%
Liraglutide (Saxenda)	GLP-1 RA	~5–8%
Naltrexone/bupropion ER (Contrave)	Opioid antagonist + NE/DA reuptake inhibitor (POMC activation in hypothalamus)	~5–6%
Phentermine	Sympathomimetic amine (↑ NE) → appetite suppression	~5% (short term)
Orlistat (Xenical, Alli OTC)	**Lipase inhibitor** → ↓ fat absorption ~30%	~3%
Setmelanotide (Imcivree)	MC4R agonist	Genetic obesity only (POMC, PCSK1, LEPR deficiency, Bardet-Biedl)

- **GLP-1-based drugs (tirzepatide, semaglutide)** = preferred first-line drugs in most guidelines (AGA 2022, Obesity Medicine) because of the greatest weight loss and cardiometabolic benefits: **semaglutide ↓ MACE** in obesity with established CVD (SELECT trial; FDA CV indication) • **tirzepatide approved for moderate–severe OSA** in obesity • semaglutide approved for **MASH** (with fibrosis).
- **T2DM + obesity:** GLP-1 RA or tirzepatide (treat both) • avoid weight-gaining diabetes drugs.
- **Metformin**, SGLT2 inhibitors, and topiramate alone are not FDA-approved for obesity • **herbal supplements are not recommended**.
- **Pediatrics (≥ 12 years):** semaglutide, liraglutide, phentermine/topiramate, orlistat • phentermine > 16 years.

GLP-1 receptor agonists and tirzepatide — key points

- **Dose titration** every 4 weeks to reduce GI effects (nausea, vomiting, diarrhea, constipation — most common; usually during titration).
- **Boxed warning: thyroid C-cell tumors → contraindicated with personal/family history of medullary thyroid carcinoma (MTC) or MEN 2**.
- **Other ADRs: pancreatitis** (stop if suspected), **gallbladder disease** (cholelithiasis), **AKI** (dehydration from GI losses), **hypoglycemia** with insulin/sulfonylureas (↓ their doses), ↑ heart rate, diabetic retinopathy complications (rapid glucose lowering — semaglutide), suicidal ideation monitoring, injection-site reactions, risk of **pulmonary aspiration** during anesthesia (delayed gastric emptying), hair loss, muscle loss.
- **Delayed gastric emptying** → may alter absorption of oral drugs.
- **Tirzepatide ↓ oral contraceptive efficacy** → use non-oral contraception or add a barrier method for **4 weeks after starting and after each dose increase**.
- **Pregnancy:** stop **semaglutide at least 2 months before planned conception** (long half-life).
- Missed dose: semaglutide — take within 5 days (otherwise skip) • tirzepatide — within 4 days • if ≥ 2 weeks missed → consider restarting titration.
- Avoid duplicating with other GLP-1 RA products (e.g., Ozempic/Mounjaro for diabetes are the same molecules).

Phentermine and phentermine/topiramate

- **Phentermine:** 15–37.5 mg every morning (or 8 mg TID — Lomaira) • **C-IV** • approved for **short-term use (≤ 12 weeks)** (often used longer off-label) • ADRs: **↑ BP and HR**, palpitations, **insomnia** (take in the morning), dry mouth, nervousness, constipation • **CI: CVD** (CAD, stroke, arrhythmia, HF), **uncontrolled HTN**, **hyperthyroidism**, **glaucoma**, agitated states, **history of drug abuse**, **MAOIs within 14 days**, pregnancy, breastfeeding.
- **Phentermine/topiramate ER (Qsymia): 3.75/23 mg daily × 14 days → 7.5/46 mg** (every morning) • if < 3% loss at 12 weeks → ↑ to 11.25/69 for 14 days → 15/92 mg • if < 5% loss after 12 weeks on 15/92 → **stop by tapering** (every other day for ≥ 1 week — **seizure risk** if stopped abruptly).
- **Qsymia REMS — teratogenic** (topiramate → **oral clefts**) → **pregnancy test before starting and monthly** + effective contraception.
- **Topiramate ADRs: paresthesia**, **dysgeusia** (altered taste — carbonated drinks), **cognitive impairment** (word-finding, concentration), **kidney stones**, **metabolic acidosis** (↓ bicarbonate), **acute angle-closure glaucoma**, oligohidrosis/hyperthermia, depression/suicidality, ↑ HR, hypokalemia with non-K-sparing diuretics.

Naltrexone/bupropion ER (Contrave)

- **Titration (8/90 mg tablets):** week 1: 1 tab AM • week 2: 1 AM + 1 PM • week 3: 2 AM + 1 PM • **week 4 onward: 2 tabs BID (32/360 mg/day)** • stop if < 5% at 12 weeks on maintenance.
- **Boxed warning: suicidal thoughts/behavior** (bupropion; antidepressant) • neuropsychiatric effects.
- **Seizures** (dose-related) → **CI: seizure disorder, bulimia/anorexia nervosa**, **abrupt discontinuation of alcohol, benzodiazepines, barbiturates, antiepileptics • do not take with a high-fat meal** (↑ levels → seizures) • max 2 tabs per dose.
- **CI: chronic opioid use** (naltrexone → precipitated withdrawal; opioids ineffective) • **uncontrolled HTN** • **MAOIs within 14 days** • pregnancy.
- ADRs: **nausea** (most common), constipation, headache, vomiting, dizziness, **insomnia**, dry mouth, ↑ BP/HR, hepatotoxicity (rare).
- **DDI: bupropion = strong CYP2D6 inhibitor** → ↑ metoprolol, TCAs, SSRIs, antipsychotics, ↓ tamoxifen/codeine activation • bupropion is a **CYP2B6 substrate** (ticlopidine, clopidogrel ↑ levels) • lowers seizure threshold with other drugs.
- Good choice for patients who also want **smoking cessation** or have depression (not with alcohol withdrawal).

Orlistat

- **120 mg TID with each fat-containing meal** (during or up to 1 h after) • OTC Alli 60 mg TID • skip dose if meal skipped or contains no fat.
- **GI ADRs: oily spotting**, **flatus with discharge**, **fecal urgency/incontinence**, fatty stools → worse with high-fat meals (**fat ≤ 30% of calories**).
- **↓ fat-soluble vitamins (A, D, E, K) and beta-carotene → multivitamin once daily at bedtime** (≥ 2 h before/after orlistat).
- Rare: severe liver injury, **oxalate nephrolithiasis/nephropathy**, pancreatitis, cholelithiasis.
- **CI: chronic malabsorption syndrome**, **cholestasis**, pregnancy.
- **DDIs: levothyroxine** (separate ≥ 4 h → hypothyroidism) • **cyclosporine** (separate ≥ 3 h, ↓ levels) • **warfarin** (↓ vitamin K → ↑ INR) • **antiepileptics** (↓ absorption → seizures) • amiodarone (↓ absorption) • antiretrovirals (loss of virologic control).

Bariatric (metabolic) surgery

- **Indications (ASMBS/IFSO 2022): BMI ≥ 35** regardless of comorbidity • **BMI 30–34.9 with metabolic disease** (T2DM) • Asian: BMI ≥ 27.5 • (older criteria: BMI ≥ 40, or ≥ 35 with comorbidity).
- Procedures: **sleeve gastrectomy** (most common, restrictive), **Roux-en-Y gastric bypass** (restrictive + malabsorptive), adjustable band (less used).
- **Most effective long-term** weight loss (25–30%), T2DM remission.
- **Post-op medication issues: avoid extended-release/enteric-coated** products (use IR, liquids, chewable, crushed) • **avoid NSAIDs** (marginal ulcers) • ↓ absorption of oral contraceptives (bypass → non-oral contraception) • ↑ **alcohol sensitivity** (alcohol use disorder risk) • adjust diabetes/antihypertensive drugs (↓ doses, hypoglycemia risk) • ursodiol for 6 months to prevent

gallstones.

- **Lifelong supplements:** multivitamin, **vitamin B12**, **iron**, **calcium citrate** (1,200–1,500 mg), **vitamin D** (3,000 IU), thiamine, folate • complications: dumping syndrome, deficiencies (anemia, **Wernicke's**, osteoporosis).

12. Obesity Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Tirzepatide (Zepbound)	**2.5 mg SC weekly × 4 weeks** → ↑ by 2.5 mg every 4 weeks → **5, 10, or 15 mg weekly**	N/V/D, constipation, pancreatitis, gallbladder disease, AKI	**Boxed: MTC/MEN2** • ↓ **oral contraceptives** (4 weeks after start/each increase) • OSA indication
Semaglutide (Wegovy)	**0.25 mg weekly** → 0.5 → 1 → 1.7 → **2.4 mg weekly** (every 4 weeks)	Same + retinopathy complications, ↑ HR	**Boxed: MTC/MEN2** • ↓ MACE (CV indication) • stop 2 months before pregnancy • ↓ insulin/SU doses
Liraglutide (Saxenda)	**0.6 mg SC daily** → ↑ 0.6 mg weekly → **3 mg daily**	Same as GLP-1 RA	Boxed: MTC/MEN2 • ≥ 12 years • daily injection
Phentermine	**15–37.5 mg every morning**	↑ BP/HR, insomnia, dry mouth, anxiety	**C-IV** • ≤ 12 weeks • CI CVD, uncontrolled HTN, hyperthyroidism, glaucoma, **MAOIs**, abuse history
Phentermine/topiramate ER (Qsymia)	3.75/23 mg × 14 days → **7.5/46 mg** → max 15/92 mg	**Paresthesia, dysgeusia**, insomnia, cognitive effects, **kidney stones**, metabolic acidosis, glaucoma	**REMS — teratogenic** (monthly pregnancy test) • **taper when stopping** (seizures) • C-IV
Naltrexone/bupropion ER (Contrave)	Titrate weekly to **2 tabs BID (32/360 mg)**	**Nausea**, constipation, headache, insomnia, ↑ BP, **seizures**	**Boxed: suicidality** • CI **opioids**, seizures, eating disorders, alcohol/BZD withdrawal, MAOIs • **CYP2D6 inhibitor** • no high-fat meals
Orlistat	**120 mg TID with fat-containing meals** (OTC 60 mg)	**Oily spotting, flatus with discharge, fecal urgency**	**MVI at bedtime** • separate **levothyroxine 4 h**, cyclosporine 3 h • ↑ INR warfarin • ↓ antiepileptics • CI malabsorption, cholestasis
Setmelanotide	2–3 mg SC daily	**Skin hyperpigmentation**, injection-site reactions, depression, sexual adverse effects	Genetic obesity only

Exam Traps and Frequently Tested Points

1. **TSH = best screening test** for primary thyroid disease • **central hypothyroidism → follow free T4**, not TSH.
2. **Hashimoto's → anti-TPO • Graves' → TSI/TRAb • RAIU high** in Graves'/toxic nodules, **low** in thyroiditis and exogenous hormone.
3. **Levothyroxine = 1.6 mcg/kg/day • elderly/CAD start 12.5–25 mcg** • check TSH every **4–8 weeks** • half-life 7 days.
4. Levothyroxine on an **empty stomach 30–60 min before breakfast • separate 4 h** from Ca, Fe, antacids, sucralfate, bile acid sequestrants, orlistat.
5. **Pregnancy: ↑ levothyroxine ~30% (2 extra doses/week)** immediately • goal TSH < 2.5.
6. Estrogen and pregnancy ↑ **TBG** → ↑ dose • enzyme inducers ↑ dose • correcting hypothyroidism ↑ **warfarin** effect.
7. **Myxedema coma:** IV levothyroxine 200–400 mcg + **hydrocortisone 100 mg q8h first/together** • passive rewarming.

8. **Methimazole = preferred** thionamide • **PTU: 1st trimester and thyroid storm • PTU boxed hepatotoxicity** • MMI teratogenic.
9. **Agranulocytosis** with thionamides → **fever or sore throat → stop the drug and get a CBC**.
10. Thionamides take **4–8 weeks** to work (stored hormone) → **beta-blocker** for symptoms • propranolol ↓ T4→T3.
11. **Thyroid storm order: PTU → (≥ 1 h later) iodine → propranolol → hydrocortisone • acetaminophen, not aspirin**.
12. **RAI:** CI in pregnancy/breastfeeding • → hypothyroidism • can worsen ophthalmopathy (prednisone) • stop MMI 3–5 days before.
13. **Thyroiditis (postpartum, painless, subacute)** → **beta-blocker only** (no thionamides).
14. **Amiodarone:** hypo- or hyperthyroidism • **type 1 → thionamide • type 2 → prednisone** • TFTs every 3–6 months • **lithium** → hypothyroidism.
15. **Nonthyroidal illness** (↓ T3) → do not treat • **biotin** → false lab results.
16. **Osteoporosis: T-score ≤ −2.5** • osteopenia −1 to −2.5 • **hip/vertebral fragility fracture = osteoporosis** regardless of BMD.
17. **Z-score** for premenopausal women and men < 50 • screen **women ≥ 65**, men ≥ 70.
18. Treat osteopenia if **FRAX ≥ 20% major or ≥ 3% hip**.
19. **Calcium 1,200 mg/day** (women > 50) • **max 500–600 mg per dose • carbonate 40% with meals • citrate 21% any time (PPIs)**.
20. **Vitamin D 800–1,000 IU/day** • goal ≥ 30 ng/mL • correct hypocalcemia/vitamin D before bisphosphonates and denosumab.
21. **Glucocorticoids** = most common drug cause • also PPIs, anticonvulsants, heparin, **DMPA (boxed, 2 years)**, aromatase inhibitors, excess levothyroxine.
22. **Oral bisphosphonates:** morning, empty stomach, **full glass of plain water**, **upright 30 min** (ibandronate 60), no food/meds 30 min.
23. **Ibandronate and raloxifene → vertebral fractures only** • alendronate, risedronate, zoledronic acid, denosumab → hip too.
24. **Zoledronic acid 5 mg yearly** • CI CrCl < 35 • acute-phase reaction • **drug holiday** after 5 y oral / 3 y IV if not high risk.
25. **ONJ** (dental exam first) and **atypical femur fracture** (thigh pain) with bisphosphonates and denosumab.
26. **Denosumab 60 mg SC q6 months • no holiday — rebound vertebral fractures** → switch to bisphosphonate • **hypocalcemia (boxed in CKD)**.
27. **Very high risk → anabolic first** (teriparatide 20 mcg, abaloparatide 80 mcg, romosozumab) → then antiresorptive.
28. **Teriparatide:** orthostatic hypotension, hypercalcemia • avoid in Paget's, radiation, open epiphyses.
29. **Romosozumab:** 12 months only • **boxed MI/stroke** — not within 1 year of MI/stroke.
30. **Raloxifene:** ↓ breast cancer, **boxed VTE/stroke**, hot flashes • stop 72 h before immobilization • **calcitonin** = least effective, malignancy risk, analgesic.
31. **BMI ≥ 30 = obesity** (≥ 25 in Asians) • waist > 40 in men, > 35 in women • goal **5–10%** loss.
32. **Drug therapy: BMI ≥ 30 or ≥ 27 + comorbidity • stop if < 5% loss at 12–16 weeks** • all CI in pregnancy.
33. **Tirzepatide (~20%) > semaglutide (~15%)** > Qsymia > liraglutide/Contrave > orlistat • GLP-1s: **boxed MTC/MEN2**, pancreatitis, gallstones.
34. **Tirzepatide ↓ oral contraceptive efficacy** (4 weeks after start/increase) • stop semaglutide **2 months before pregnancy**.
35. **Phentermine:** C-IV, short term, CI CVD/HTN/hyperthyroidism/glaucoma/MAOIs • **Qsymia: REMS (oral clefts), taper off, paresthesia, kidney stones**.
36. **Contrave: CI opioids, seizures, eating disorders** • boxed suicidality • CYP2D6 inhibitor • no high-fat meals.
37. **Orlistat:** TID with fatty meals • oily spotting • **MVI at bedtime** • separate levothyroxine 4 h, cyclosporine 3 h • ↑ INR.
38. **Weight-gaining drugs:** olanzapine, clozapine, mirtazapine, paroxetine, valproate, gabapentin, insulin, SU, TZDs, steroids, DMPA → review first.

39. **Bariatric surgery:** BMI ≥ 35 (or 30 with T2DM) • avoid **ER products and NSAIDs** • lifelong B12, iron, calcium citrate, vitamin D.

> **Note:** doses and cut-offs follow current guidelines (ATA, AACE 2020, BHOF, ACR 2022, ACP 2023, AGA 2022). If your course slides or reference use slightly different numbers, **use your course numbers for the exam**.